Jiayu Yuan

Wuhan, China | jiayuyuan52@gmail.com

Education

Huazhong University of Science and Technology — Wuhan, China
Bachelor of Computer Science — *Sep. 2020 – Jun. 2024*
- Weighted Average Score: 87.8 / 100 — GPA: 3.88 / 4.0

Huazhong University of Science and Technology — Wuhan, China
Master of Computer Science — *Sep. 2024 – Present*
- Weighted Average Score: 87.52 / 100

Publications

AdaptiveIMI: Efficient KV Cache Management for Long-Context LLM Inference on Limited GPU Resources

- **First Author**
- Submitted to *VLDB 2027*
- Proposes an efficient KV cache management mechanism for long-context LLM inference under limited GPU memory.

An Experimental Evaluation of Hybrid Querying on Vectors

- **Second Author**
- *VLDB 2026*
- Provides a systematic evaluation of state-of-the-art hybrid querying methods.

Research Experience

AdaptiveIMI: KV Cache Optimization for LLM Inference

- Designed a KV cache management system for long-context LLM inference under GPU memory constraints.
- Developed an IMI-based clustering and retrieval mechanism for efficient attention approximation.
- Improved inference efficiency and reduced GPU memory usage compared to full attention.

Hybrid Querying in Vector Databases

- Conducted a systematic study of ANN search methods under hybrid query scenarios.
- Designed benchmark datasets and experiments for comprehensive evaluation.
- Performed large-scale experiments to analyze performance trade-offs across methods and systems.
- Summarized key insights and practical guidelines for hybrid query processing.

Research Interests

- Efficient LLM inference systems, including KV cache management, memory optimization, and resource-efficient execution.
- ANN search, vector retrieval, and RAG for improving efficiency and scalability in LLM inference.
- Database and data management techniques for large-scale systems, including indexing, efficient query processing, and storage optimization.

Honors & Awards

First-Class Academic Scholarship — 2024–2025

Second-Class Academic Scholarship — 2025–2026

Technical Skills

Programming: Python, C/C++, CUDA
Frameworks: PyTorch, FAISS
Systems: Linux, GPU Programming, LLM Inference Optimization

Huazhong University of Science and Technology
Add: 1037#Luoyu Rd., Wuhan, China 430074
華中科技大學
中华人民共和国 湖北 武汉

Academic Transcript

Name: Yuan Jiayu
Student ID: M202474096
College: School of Computer Science and Technology
Major: Computer Science and Technology

Date of Enrollment: September, 2024
Date of Printing: April, 2026
Academic Program: Master's Program

NO.	Courses Name	Hours	Credit	Score	Duration of Study
1	IT China (Introduction to Information Frontier)	16	1.0	91	Autumn of 2024
2	First Foreign Language(English I)	32	2.0	exemption	Autumn of 2024
3	Design and Practice of Distributed Storage System	32	2.0	94	Autumn of 2024
4	Advanced Computer Architecture Seminar	32	2.0	80	Autumn of 2024
5	Advanced algorithm design and analysis	32	2.0	79	Autumn of 2024
6	Matrix Theory	48	3.0	84	Autumn of 2024
7	Reinforcement learning	32	2.0	92	Autumn of 2024
8	Artificial Intelligence	32	2.0	97	Autumn of 2024
9	Theory and Technology of Information Storage	32	2.0	88	Autumn of 2024
10	Topics on Resources and Environmental Economics	32	2.0	83	Autumn of 2024
11	Dialectics of Nature	18	1.0	83	Autumn of 2024
12	Thesis writing	16	1.0	90	Spring of 2025
13	A study on the theory and practice of socialism with Chinese characteristics	36	2.0	91	Spring of 2025
14	Stress and Coping	16	1.0	89	Spring of 2025
15	Thesis proposal	0	1.0	pass	Autumn of 2025

Total credits: 26.00 **Courses weighted average score: 87.52**

Remarks: Two grading systems we employ are as follows:

1. The Percentage System: 60 is passing, 100 is full mark;

2. Two-Degree Grading: pass or fail.

3. Courses weighted average score = $\frac{\sum(\text{Degree Course Credits} * \text{grade})}{\sum \text{Degree Course Credits}}$ **, only count marks of the Percentage System.**

Graduate School

Dean:

Huazhong University of Science and Technology

華中科技大學
HUAZHONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

UNDERGRADUATE ACADEMIC RECORD

Name: Yuan Jiayu
Student ID: U202015599
Department: School of Computer Science and Technology
Major: Computer Science and Technology
Date of Entrance: 01/09/2020
Length of Schooling: 4 years

Course	Credit	Result
2020-2021 1st Semester		
Advanced Programming Language (C)	3.0	83
Advanced Programming Language Experiments	1.0	85
Computational Thinking	2.0	81
Military Training	1.0	89
Morals, Ethics and Fundamentals of Law	2.5	85
Calculus (I) (A)	5.5	99
Linear Algebra	2.5	88
Computer Skills Practice for Freshman	1.0	91
Introduction to Information Technology	1.5	93
Yoga (I)	1.0	89
Chinese	2.0	90
Comprehensive English (I)	3.5	88
2020-2021 2nd Semester		
Physics (I)	4.0	87
Probability Theory and Mathematical Statistics	2.5	92
Engineering Training (VII)	1.0	96
Military Theory	1.0	96
Discrete Mathematics (I)	3.5	74
Data Structure	3.0	80
Data Structure Experiments	1.0	90
Ideological and Political Course Social Practice	0.0	B
Calculus (I) (B)	5.5	88
Experiments of Physics(I)	1.0	80
Psychology & Life	2.0	92
Yoga (II)	1.0	93
Survey of Modern Chinese History	2.5	88
Comprehensive English (II)	3.5	87
2021-2022 1st Semester		
Course Project of Programming	1.0	86
Big Data Introduction	1.5	83
Physics (II)	4.0	92
Circuit Theory (V)	4.0	85
Complex Function and Integral Transform	2.5	96
Principles of Functional Programming	2.0	86
Discrete Mathematics (II)	1.5	94
Introduction to Basic Principles of Marxism	2.5	98
Studies on Historical Figures of Republic of China (1912–49)	2.0	88
Introduction to Artificial Intelligence	1.5	84
Digital Circuit and Logic Design (I)	3.0	94
Digital Circuit and Logic Design Experiments	1.0	90
Tennis(level 1)	1.0	89
Experiments of Physics(II)	0.8	85

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Course	Credit	Result
Chinese Cultural History	2.0	94
2021-2022 2nd Semester		
Advanced Programming Language (JAVA)	2.5	76
Advanced Programming Language Experiments (JAVA)	0.8	97
Big Data Processing	2.5	96
Assembly Language Programming	1.5	82
Assembly Language Programming Experiments	1.0	84
Foundation of Computer Systems	2.5	81
Computer Organization	3.0	81
Computer Organization Experiments	0.5	85
General Introduction to Mao Zedong Thought and Socialist Theory with Chinese Characteristics	4.5	80
Analog Electronic Technology(II)	3.0	93
Design and Analysis of Algorithms	2.0	95
Tennis(level 2)	1.0	99
Signal and Linear System	2.0	86
2022-2023 1st Semester		
Operating Systems	3.0	94
Operating Systems Experiments	0.5	89
Big Data Analysis	2.5	91
Critical Thinking of College Students	2.0	96
Computer Telecommunications and Network	2.5	85
Computer Telecommunications and Network Experiments	1.0	94
Software Engineering	2.0	88
Market Research(General Elective)	2.0	91
Database Systems	3.0	86
Database Systems Experiments	1.0	94
Numerical Analysis	2.0	84
Project of Hardware System	1.0	67
2022-2023 2nd Semester		
Compiler Principles	2.0	91
Compiler Principles Experiments	1.0	98
Course Project of Operating Systems	1.0	87
Computer Architecture	2.0	91
Situation and Policy	2.0	91
2023-2024 1st Semester		
Engineering Internship	1.5	84
Comprehensive Training of System Ability	2.0	91
2023-2024 2nd Semester		
Undergraduate Thesis	7.0	84

Credits:146.0 Cumulative Average Grade:87.8
GPA: 3.88

Provost: 周平良

Undergraduate College
Huazhong University of Science and Technology

华中科技大学本科生院
本科成绩专用章

Issue Date:3/4/2026

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